

VISHVANATH MADASAMY

Systems Engineer | Infrastructure and Platform Specialist

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PROFESSIONAL SUMMARY

Results-driven Systems Engineer with expertise in service cost optimization, root cause analysis (RCA), and design-to-production engineering. Proven ability diagnosing complex hardware/software systems through log analysis and data-driven investigations. Skilled in RCA, business case development, and translating engineering feedback into actionable product specifications. AWS Certified AI Practitioner & Cloud Practitioner with hands-on experience in process improvement, quality assurance automation, and cost reduction (45% optimization demonstrated).

CERTIFICATIONS

AWS Certified AI Practitioner (AIF-01)

Amazon Web Services

Active: Mar 2026 – Mar 2029

AWS Certified Cloud Practitioner (CLF-02)

Amazon Web Services

Active: Mar 2026 – Mar 2029

TECHNICAL SKILLS

Product Engineering & Service Design: Service Cost Analysis, Root Cause Analysis (RCA), Engineering Specifications, Quality Control Automation, Product Requirements Definition, Service Efficiency Optimization, Case Studies, Cost Reduction Analysis

Systems & Troubleshooting: Complex System Troubleshooting, Hardware/Software Integration, System Diagnostic Tools, Log Analysis, Failure Mode Mitigation, Systems Approach Engineering, EE/ME Fundamentals, Network Systems, Connected Hardware, IoT Systems

Programming & Data Analysis: Python (Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow), SQL, JavaScript, TypeScript, Data Extraction, Complex Data Manipulation, ETL Pipelines, Data-Driven Analysis

Cloud & Infrastructure: AWS (SageMaker, Bedrock), GCP, Docker, Kubernetes, CI/CD Pipelines, Ansible/Chef, OpenSearch, Cloud Architecture

AI & Analytics: ML-based Anomaly Detection, LLM Orchestration (LangChain), AI/Tool Integration, Power BI, Tableau, d3.js, ML Forecasting, Business Intelligence

Databases: MySQL, PostgreSQL, SQL Server, NoSQL (MongoDB), OpenSearch, Query Optimization, Schema Design, Real-time Reporting

Business & Program Management: Business Case Development, Project Ownership, Cross-functional Collaboration, ROI Analysis, Program Management, Executive Reporting, Multi-project Prioritization, Process Definition, Standardization

Soft Skills: Root Cause Analysis, Problem-Solving, Technical Communication, Cross-disciplinary Collaboration, Process Improvement, Cost Optimization, Feature Development

WORK EXPERIENCE

University of Massachusetts Dartmouth

North Dartmouth, MA

System Engineer

Sep 2025 – Present

- Architected enterprise backend systems integrating hardware/software platforms; designed Python APIs connecting INFOR POS and access control systems, enabling real-time system monitoring and cross-functional engineering feedback loops.
- Administered MySQL, PostgreSQL, OpenSearch databases with advanced schema design and query optimization; improved system reliability and real-time reporting for service operations, reducing incident response time.
- Built automated diagnostic tools using Python to parse system logs and perform ML-based anomaly detection; identified and mitigated critical failure modes through systematic RCA, enhancing operational reliability across 10,000+ users.
- Managed full lifecycle ownership of containerized systems (Docker/Kubernetes) from POC to production; drove cross-functional projects from hypothesis to validated solution.
- Developed data-driven insights from complex system behavior; translated technical findings into actionable service improvement specifications and business case analyses.

- Architected production-grade Python + SQL Server ETL pipeline; reduced operational data processing time by 45% through advanced automation and optimized query execution.
- Integrated IoT sensor hardware with Python monitoring software and designed real-time quality control mechanisms; achieved 37% improvement in system efficiency through RCA of failure modes.
- Developed REST APIs connecting hardware sensors to backend databases, enabling real-time reporting and automated QA workflows; created self-service analytics platform for cross-functional teams.
- Designed interactive Power BI/Tableau dashboards with ML forecasting; delivered data-driven insights influencing product design decisions, contributing to 30% operational efficiency improvement.
- Performed RCA on system integration issues; recommended process improvements and engineering specifications enhancing serviceability and reducing MTTR.

- Engineered serverless data monitoring and ETL pipeline for Pune Metro infrastructure using Python and ML-based alerting — achieving a 37% improvement in system efficiency and eliminating manual monitoring workflows.
- Built resilient automation framework handling real-time IoT data ingestion, system health monitoring, and automated incident reporting — enabling ops teams to detect and respond to infrastructure anomalies in real time.
- Optimized SQL queries and implemented data validation and quality checks during automated report generation — ensuring high-fidelity analytics for infrastructure stakeholders across Pune Metro's multi-building environment.

KEY PROJECTS

AI-Powered Medical Image Analysis Platform | Python, Next.js, ReactJS, Google Cloud | [GitHub](#) Oct 2025

- Built a full-stack production-grade AI analytics platform with ReactJS frontend, Python backend APIs, LLM-powered inference pipeline, and GCP cloud deployment — demonstrating end-to-end data engineering, agentic AI, and scalable cloud architecture capability.

Global Microplastics Network Visualization | Python, d3.js, Visual Analytics | [GitHub](#) May 2025

- Developed an interactive d3.js network visualization system enabling clickable node-based exploration of worldwide microplastic pollution data — implementing drill-down analytics, geographic concentration statistics, and real-time data rendering for scientific decision-making.

Car Rental Management System | Python, ReactJS, HTML/CSS, MySQL, SSMS | Database Design Coursework | [Github](#) May 2025

- Designed and developed a full-stack system with Python backend, ReactJS/HTML/CSS frontend, and MySQL/SSMS database — implementing CRUD operations, real-time availability querying, dynamic pricing logic, and an admin analytics dashboard for fleet management.

Library Assistant Chatbot — Conversational AI | Python, Rasa NLU, Flask, REST API | [Github](#) Jul 2024

- Designed and deployed an NLP agentic chatbot using Rasa NLU and Flask REST API — implementing intent classification, entity recognition, dialogue management, and webhook integration for fully automated self-service support workflows. Designed an NLP-based conversational agent for automated self-service support using intent classification and entity recognition.
- System Analysis Feature: Engineered a data module to ingest and model historical borrowing data; performed trend analysis to identify Library Checkout Trends, enabling data-driven inventory management and resource allocation.

EDUCATION

Master of Science in Data Science | GPA: 3.778 / 4.0

Aug 2024 – May 2026

University of Massachusetts Dartmouth — North Dartmouth, MA

Coursework: AI, Advanced ML, Database Design, Network Security, High Performance Scientific Computation, Data Mining, Statistical Analysis, Data Visualization

Grade: Distinction

Bachelor of Engineering — Electronics & Communication Engineering | GPA: 8.65 / 10

Jul 2020 – Apr 2024

Anna University — Chennai, India

Best Undergraduate Final Project Award 2024 — ML-based automated image detection system

Grade: First Class with Distinction.